

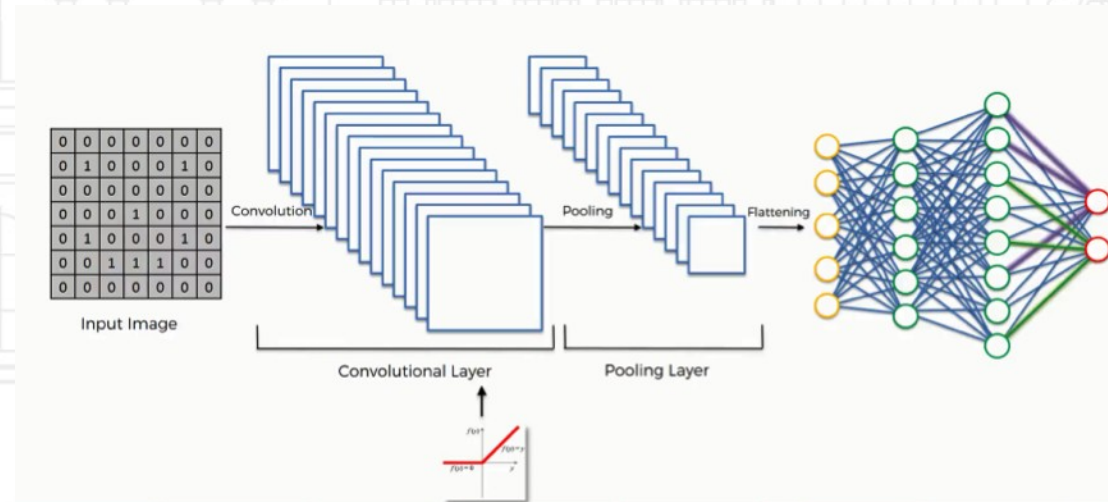


# Module 5

Convolutional Neural Networks

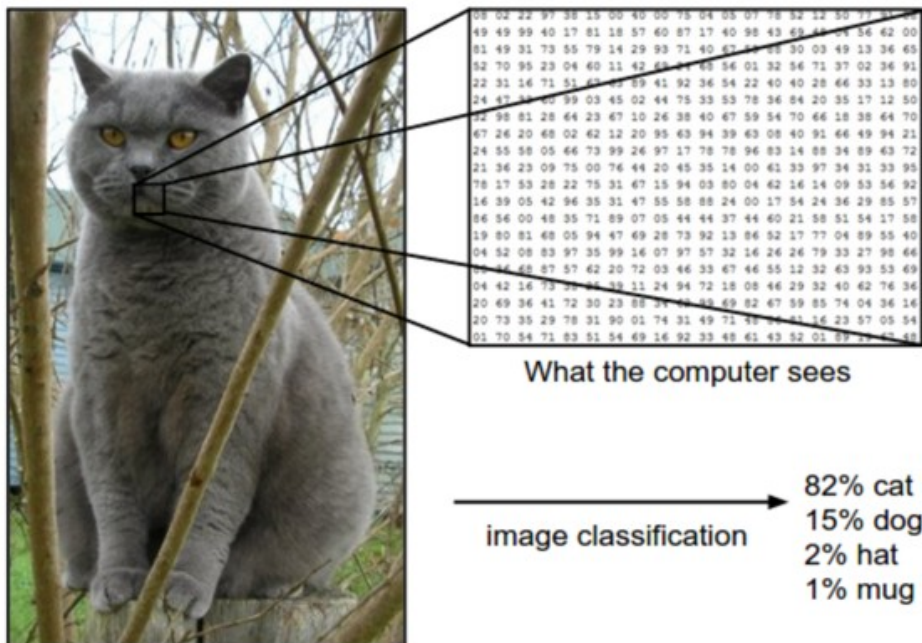


# DL: Convolutional Nets



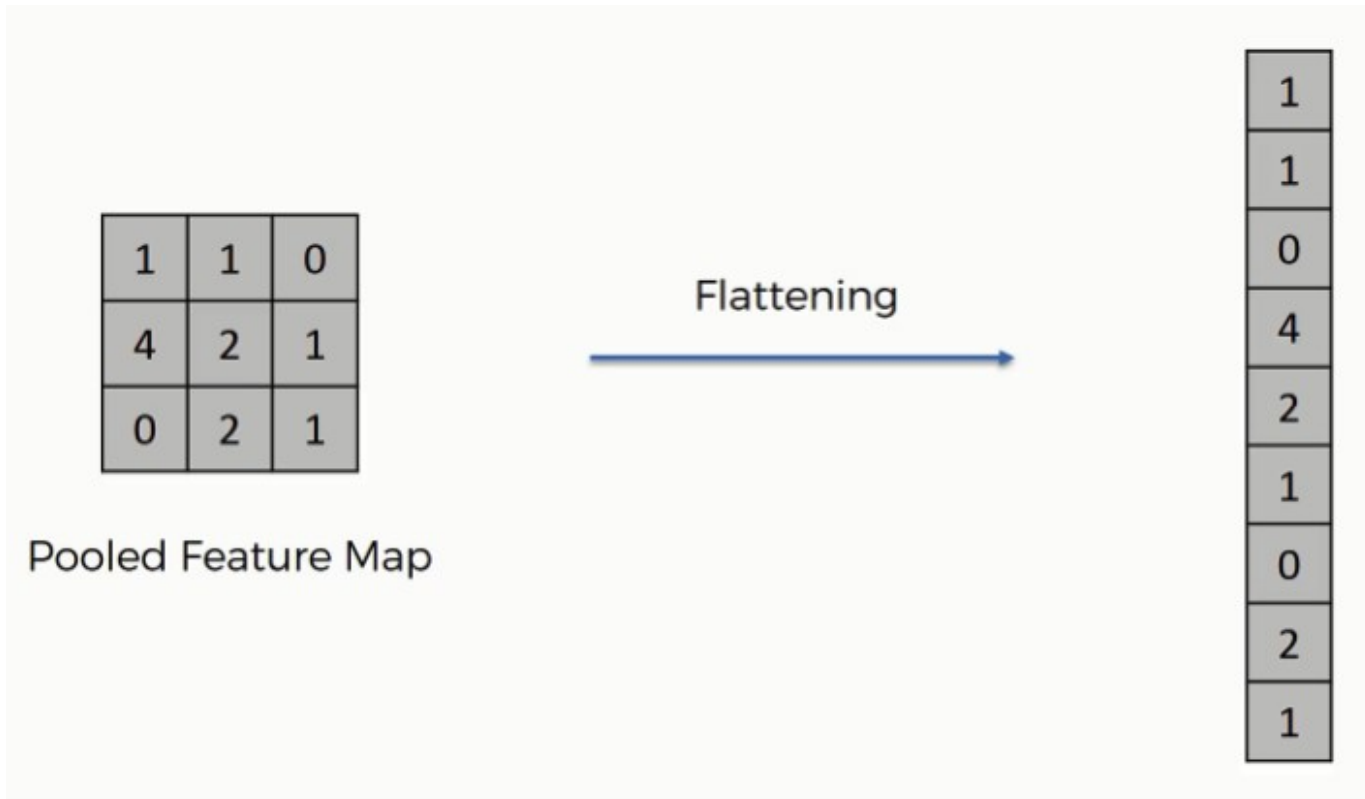
# Image Representation

- ❑ We know images are represented by tensors
- ❑ A colored pixel is represented as three separate values (RGB).
- ❑ 3 matrices (tensor) represent red (R), green (G), blue (B) channels.



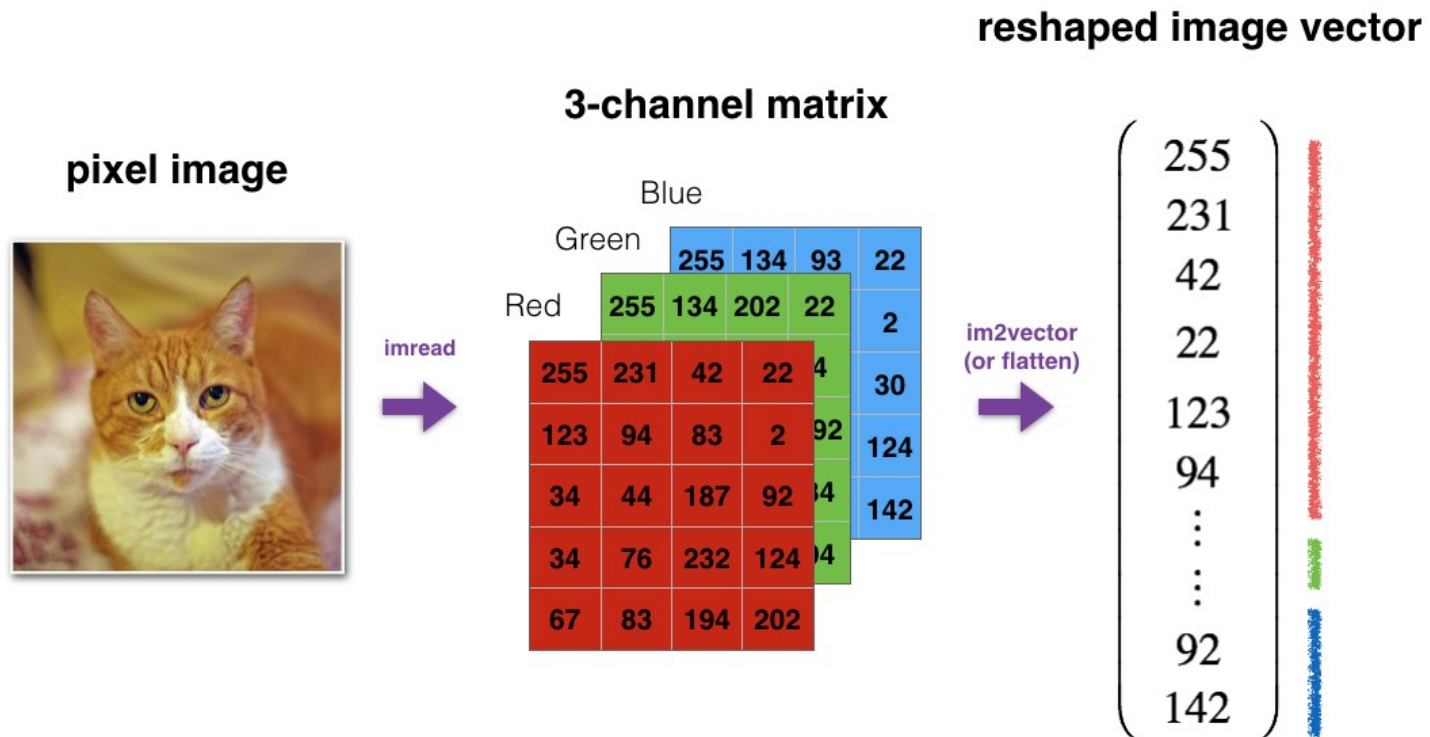
# Image Representation

- ❑ Channels can be flattened to create a single vector.



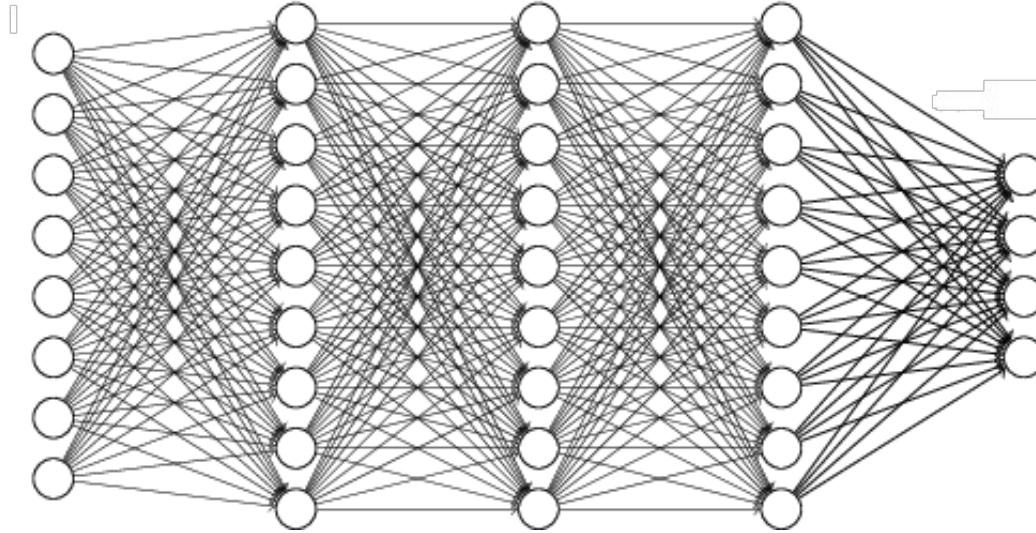
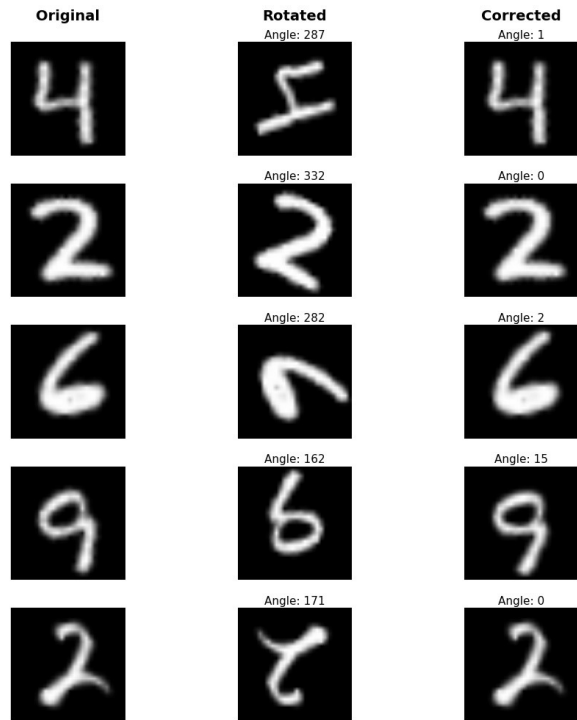
# Image Representation

- Channels can be flattened to create a single vector.



# Using Feed-Forward NNs for Images

- ❑ Feed forward NNs are not very effective with images



# Using Feed-Forward NNs for Images

## ❑ Translation Invariance



Cat



Cat



# Using Feed-Forward NNs for Images

- ❑ Other Invariances cause issues

Translation Invariance



Rotation/Viewpoint Invariance





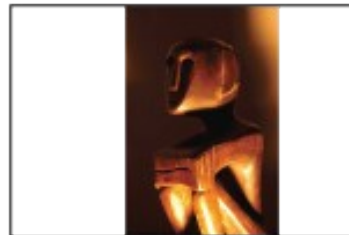
# Using Feed-Forward NNs for Images

- ❑ Other Invariances cause issues

Size Invariance

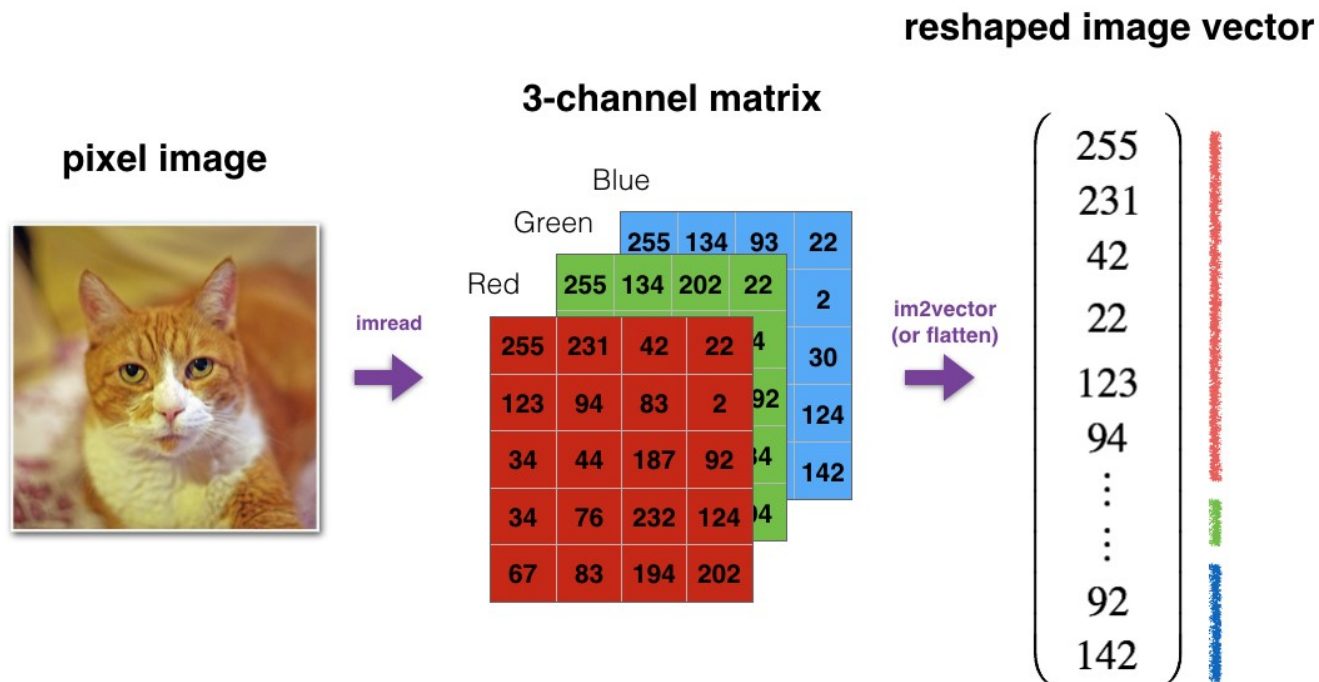


Illumination Invariance



# Using Feed-Forward NNs for Images

- ❑ Flattening tensors create an enormous amount of features.
- ❑ Predictive power declines with the added parameters estimated.





# Python



# Convolutions (think about Waldo)



# Convolutions

- A convolution (kernel, mask, filter) is a small matrix used for blurring, sharpening, embossing, edge detection

Input                      Kernel                      Output

|   |   |   |
|---|---|---|
| 0 | 1 | 2 |
| 3 | 4 | 5 |
| 6 | 7 | 8 |

\*

|   |   |
|---|---|
| 0 | 1 |
| 2 | 3 |

=

|    |    |
|----|----|
| 19 | 25 |
| 37 | 43 |

$$\begin{aligned} 0 \times 0 + 1 \times 1 + 3 \times 2 + 4 \times 3 &= 19, \\ 1 \times 0 + 2 \times 1 + 4 \times 2 + 5 \times 3 &= 25, \\ 3 \times 0 + 4 \times 1 + 6 \times 2 + 7 \times 3 &= 37, \\ 4 \times 0 + 5 \times 1 + 7 \times 2 + 8 \times 3 &= 43. \end{aligned}$$

# Convolutions

- ❑ Convolutions go over the image and produce an output

|   |   |   |   |   |
|---|---|---|---|---|
| 7 | 2 | 3 | 3 | 8 |
| 4 | 5 | 3 | 8 | 4 |
| 3 | 3 | 2 | 8 | 4 |
| 2 | 8 | 7 | 2 | 7 |
| 5 | 4 | 4 | 5 | 4 |

\*

|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
| 1 | 0 | -1 |
| 1 | 0 | -1 |

=

|   |  |  |
|---|--|--|
| 6 |  |  |
|   |  |  |
|   |  |  |

$7 \times 1 + 4 \times 1 + 3 \times 1 +$   
 $2 \times 0 + 5 \times 0 + 3 \times 0 +$   
 $3 \times -1 + 3 \times -1 + 2 \times -1$   
 $= 6$



# Convolutions

- Suppose we have an image  $I_{n \times p}$  and a kernel  $K_{k \times l}$
- The resulting image  $O_{n-k+1} \times p-l+1$

|   |   |   |   |   |
|---|---|---|---|---|
| 7 | 2 | 3 | 3 | 8 |
| 4 | 5 | 3 | 8 | 4 |
| 3 | 3 | 2 | 8 | 4 |
| 2 | 8 | 7 | 2 | 7 |
| 5 | 4 | 4 | 5 | 4 |

\*

|   |   |    |
|---|---|----|
| 1 | 0 | -1 |
| 1 | 0 | -1 |
| 1 | 0 | -1 |

=

|   |  |  |
|---|--|--|
| 6 |  |  |
|   |  |  |
|   |  |  |

$$\begin{aligned}
 &7 \times 1 + 4 \times 1 + 3 \times 1 + \\
 &2 \times 0 + 5 \times 0 + 3 \times 0 + \\
 &3 \times -1 + 3 \times -1 + 2 \times -1 \\
 &= 6
 \end{aligned}$$

$$I_{5 \times 5}$$

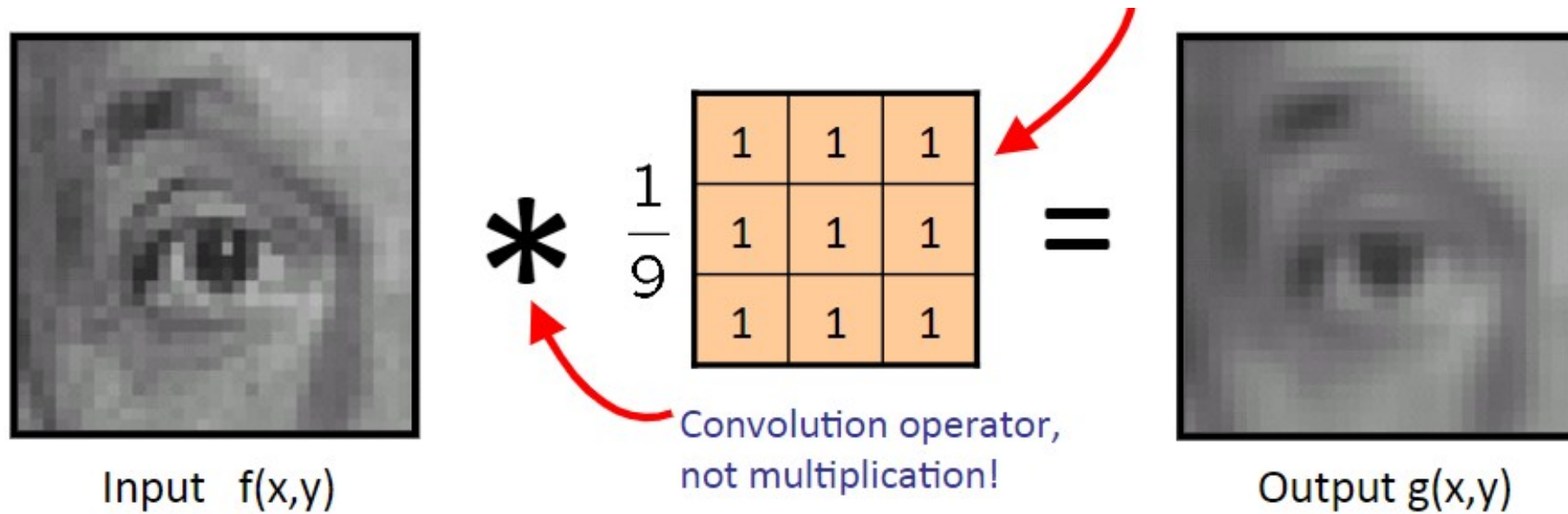
$$K_{3 \times 3}$$

$$O_{5-3+1 \times 5-3+1} = O_{3 \times 3}$$



# Convolutions Use

- ❑ Convolutions can blur (downsample) images



# Convolutions Use

- ❑ We can create multiple levels of blurring
- ❑ In other words, create multiple images of the same one.

Input Image

\*

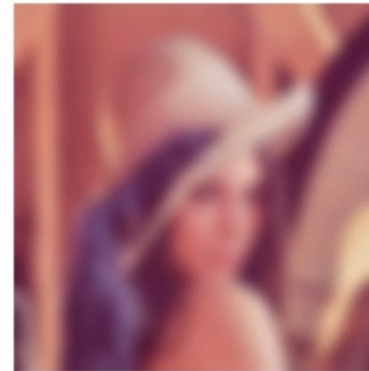
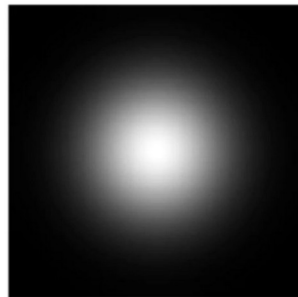
Filter (Kernel)

=

Output Image



\*

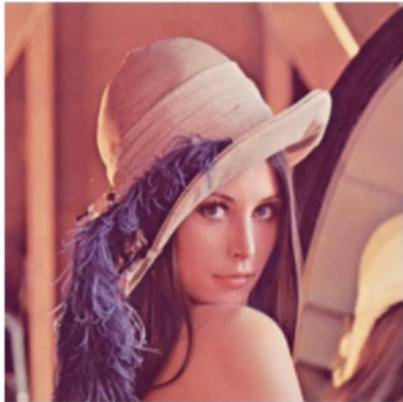


$$\begin{pmatrix} 0.0625 & 0.125 & 0.0625 \\ 0.125 & 0.25 & 0.125 \\ 0.0625 & 0.125 & 0.0625 \end{pmatrix}$$

# Convolutions Use

- We can also better detect edges.

Input Image



\*

Filter (Kernel)

=

Output Image

$$\sqrt{G_x^2 + G_y^2}$$



\*

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$
$$G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

# Convolutions Use

- ❑ Convolutions work to create new features.



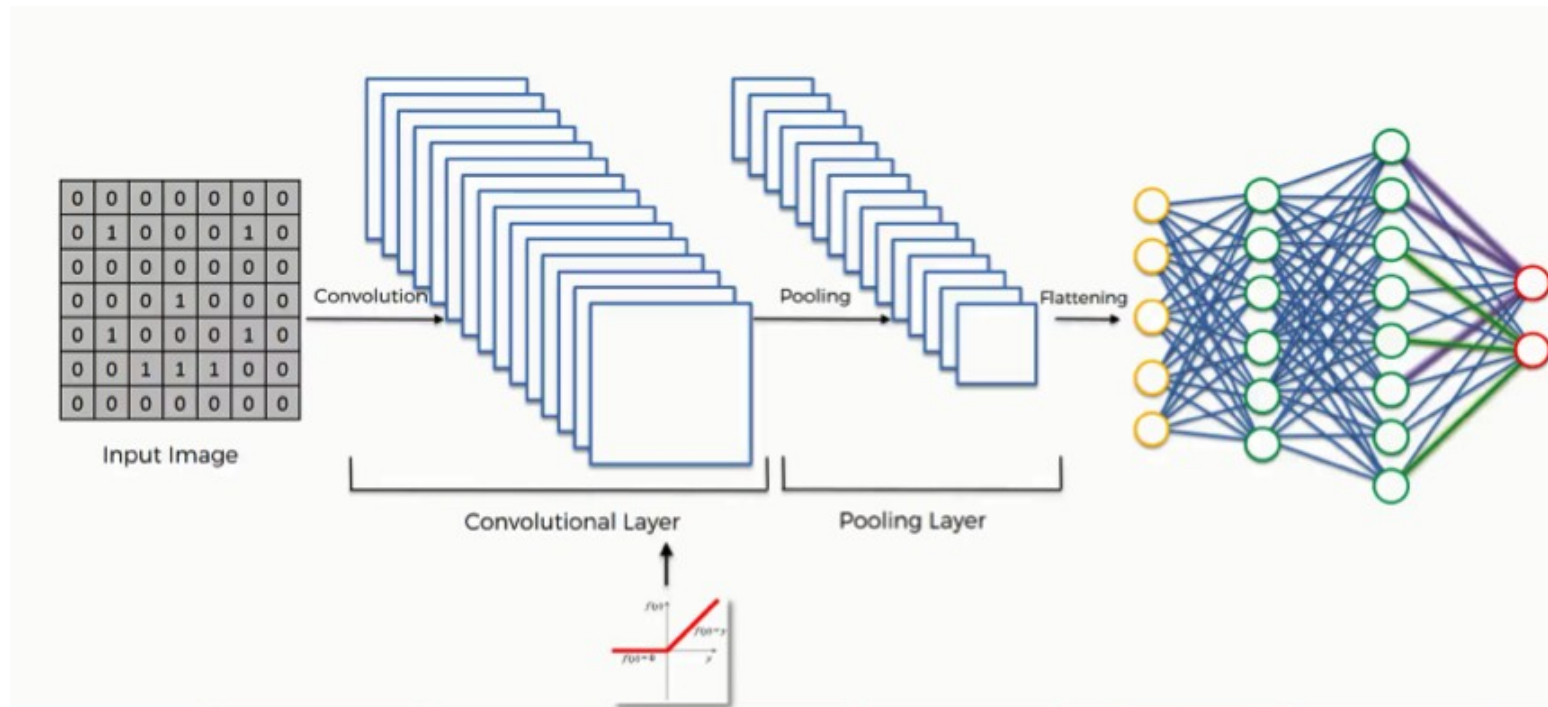
# Convolutions Use

- When we combine multiple convolutions we create a convolutional layer.



# Convolutions Use

- When we combine multiple convolutions we create a convolutional layer.





# Example

- Suppose we have the following image and convolution.

Image

|    |    |    |    |
|----|----|----|----|
| 0  | 10 | 10 | 0  |
| 20 | 30 | 30 | 20 |
| 10 | 20 | 20 | 10 |
| 0  | 5  | 5  | 0  |

\*

Filter

|   |   |
|---|---|
| 1 | 0 |
| 0 | 2 |

slido



**What is the resulting output image shape**

- ☐ Start presenting to display the poll results on this slide.

# Example

- Suppose we have the following image and convolution.

$$\begin{array}{c} \text{Image} \\ \begin{array}{|c|c|c|c|} \hline 0 & 10 & 10 & 0 \\ \hline 20 & 30 & 30 & 20 \\ \hline 10 & 20 & 20 & 10 \\ \hline 0 & 5 & 5 & 0 \\ \hline \end{array} \end{array} * \begin{array}{c} \text{Filter} \\ \begin{array}{|c|c|} \hline 1 & 0 \\ \hline 0 & 2 \\ \hline \end{array} \end{array} = \begin{array}{c} \text{Output} \\ \begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array} \end{array}$$

slido



**What is the value of X?**

☐ Start presenting to display the poll results on this slide.

# Example

- Suppose we have the following image and convolution.

| Image                                                                                                                                                                                                                         | *  | Filter | =  | Output |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------|----|--------|----|----|----|----|----|----|----|----|---|---|---|---|--|---------------------------------------------------------------------------|---|---|---|---|--|---------------------------------------------------------------------------------------------------------------------------------|----|----|--|--|--|--|--|--|--|
| <table><tr><td>0</td><td>10</td><td>10</td><td>0</td></tr><tr><td>20</td><td>30</td><td>30</td><td>20</td></tr><tr><td>10</td><td>20</td><td>20</td><td>10</td></tr><tr><td>0</td><td>5</td><td>5</td><td>0</td></tr></table> | 0  | 10     | 10 | 0      | 20 | 30 | 30 | 20 | 10 | 20 | 20 | 10 | 0 | 5 | 5 | 0 |  | <table><tr><td>1</td><td>0</td></tr><tr><td>0</td><td>2</td></tr></table> | 1 | 0 | 0 | 2 |  | <table><tr><td>60</td><td>70</td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table> | 60 | 70 |  |  |  |  |  |  |  |
| 0                                                                                                                                                                                                                             | 10 | 10     | 0  |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
| 20                                                                                                                                                                                                                            | 30 | 30     | 20 |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
| 10                                                                                                                                                                                                                            | 20 | 20     | 10 |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
| 0                                                                                                                                                                                                                             | 5  | 5      | 0  |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
| 1                                                                                                                                                                                                                             | 0  |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
| 0                                                                                                                                                                                                                             | 2  |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
| 60                                                                                                                                                                                                                            | 70 |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                               |    |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                               |    |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                 |    |    |  |  |  |  |  |  |  |

$$1*10 + 0*10 + 0*30 + 2*30 = 70$$

# Example

- Suppose we have the following image and convolution.

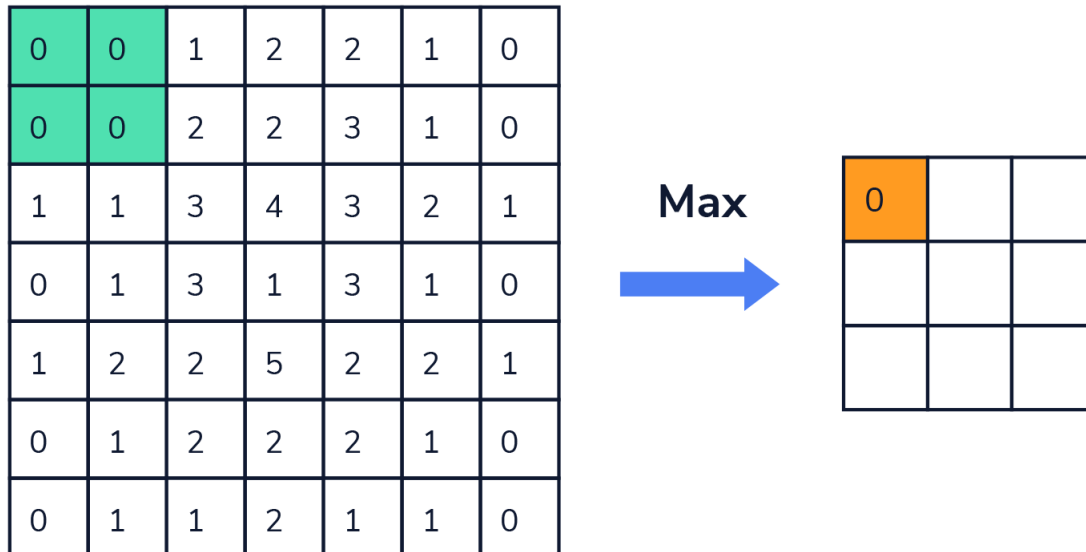
| Image                                                                                                                                                                                                                         | *  | Filter | =  | Output |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------|----|--------|----|----|----|----|----|----|----|----|---|---|---|---|--|---------------------------------------------------------------------------|---|---|---|---|--|-------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|--|--|--|--|--|
| <table><tr><td>0</td><td>10</td><td>10</td><td>0</td></tr><tr><td>20</td><td>30</td><td>30</td><td>20</td></tr><tr><td>10</td><td>20</td><td>20</td><td>10</td></tr><tr><td>0</td><td>5</td><td>5</td><td>0</td></tr></table> | 0  | 10     | 10 | 0      | 20 | 30 | 30 | 20 | 10 | 20 | 20 | 10 | 0 | 5 | 5 | 0 |  | <table><tr><td>1</td><td>0</td></tr><tr><td>0</td><td>2</td></tr></table> | 1 | 0 | 0 | 2 |  | <table><tr><td>60</td><td>70</td><td>50</td></tr><tr><td>60</td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table> | 60 | 70 | 50 | 60 |  |  |  |  |  |
| 0                                                                                                                                                                                                                             | 10 | 10     | 0  |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
| 20                                                                                                                                                                                                                            | 30 | 30     | 20 |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
| 10                                                                                                                                                                                                                            | 20 | 20     | 10 |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
| 0                                                                                                                                                                                                                             | 5  | 5      | 0  |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
| 1                                                                                                                                                                                                                             | 0  |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
| 0                                                                                                                                                                                                                             | 2  |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
| 60                                                                                                                                                                                                                            | 70 | 50     |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
| 60                                                                                                                                                                                                                            |    |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |
|                                                                                                                                                                                                                               |    |        |    |        |    |    |    |    |    |    |    |    |   |   |   |   |  |                                                                           |   |   |   |   |  |                                                                                                                                     |    |    |    |    |  |  |  |  |  |

$$1*20 + 0*30 + 0*10 + 2*20 = 60$$



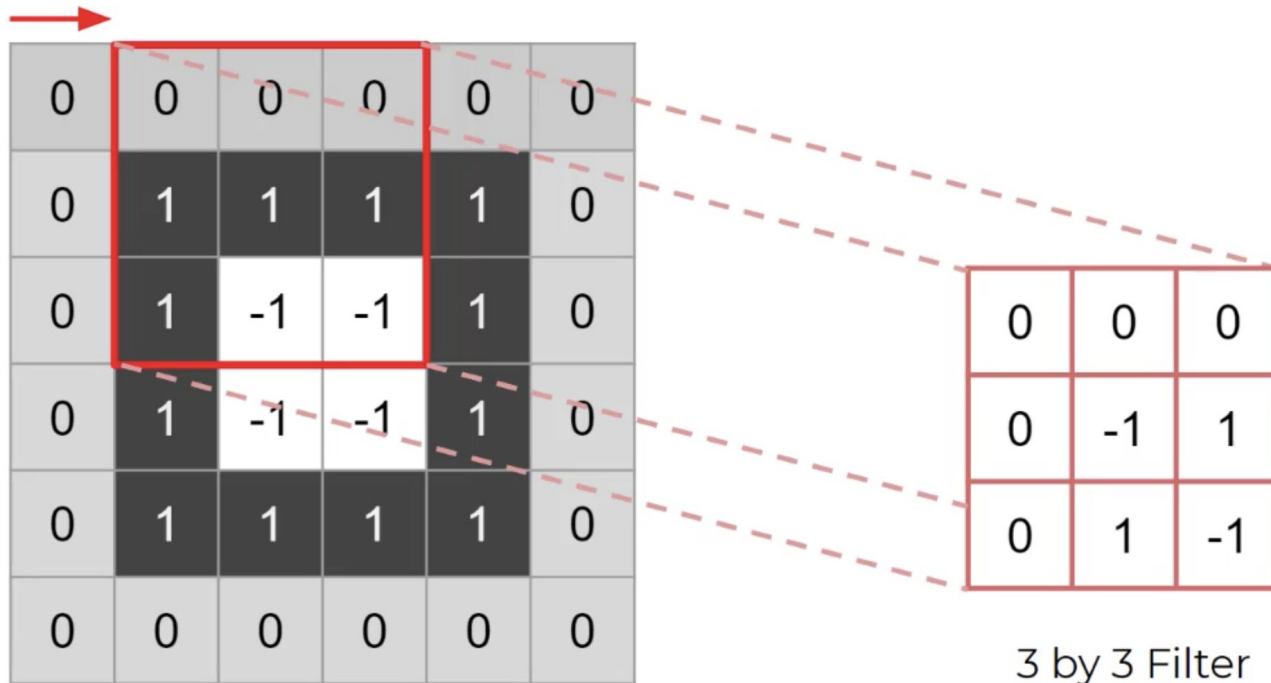
# Pooling Filter

- ❑ Pooling is also used to downsample the images
- ❑ Pooling filters keep the important parts of the images.
- ❑ Max, Min and Average Pooling Filters are the most common



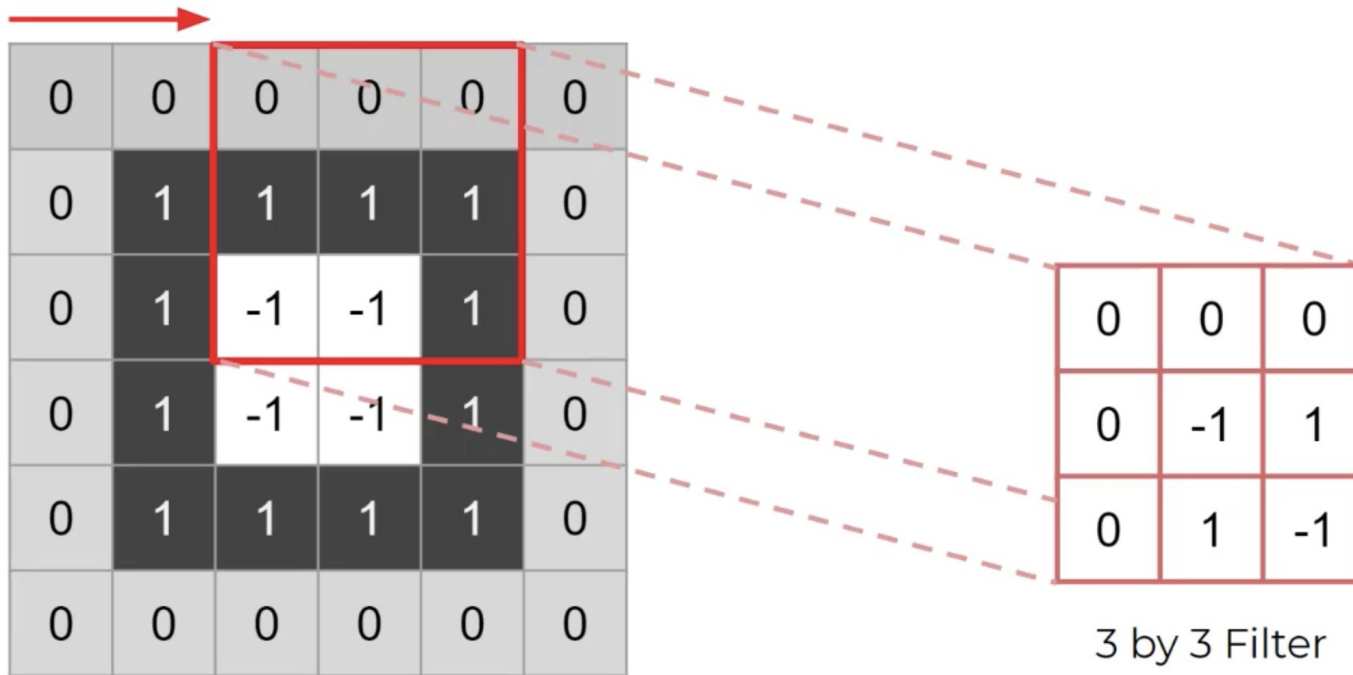
# Stride

- ❑ Dictates the movement of the kernel, or filter, across the image
- ❑ Example of (1,1) stride (most common)



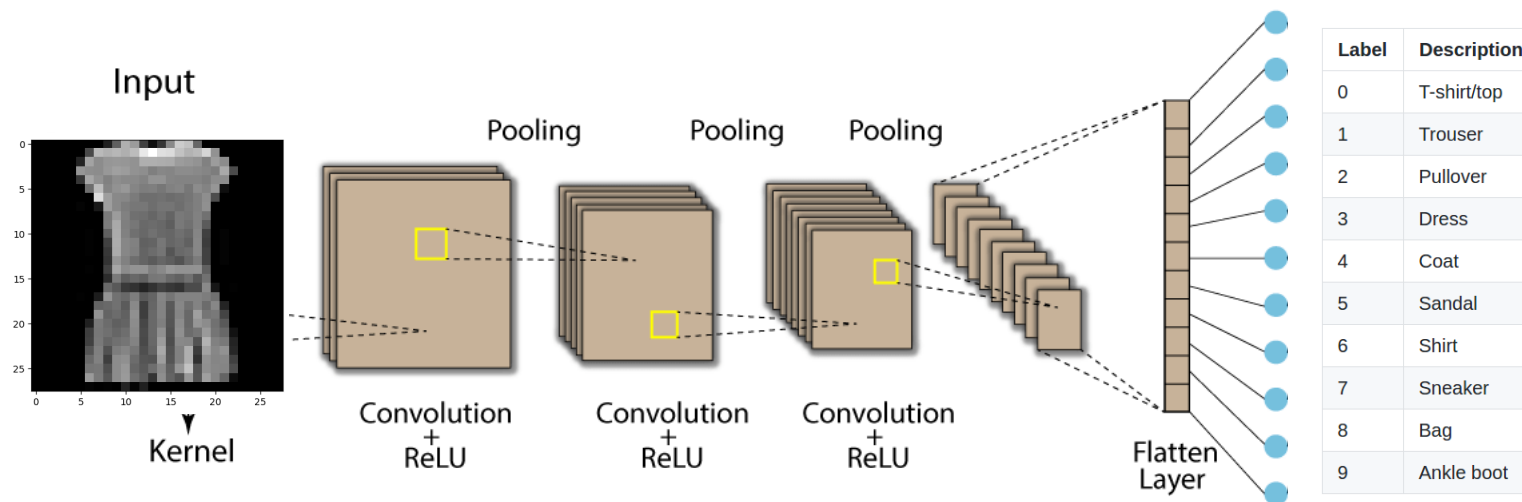
# Stride

- Example of a (2,2) stride



# Final CNN Product

- ❑ We combine multiple convolutional layers and pooling layers
- ❑ We just set the size of the kernels
- ❑ We let the network optimize values of the filters
- ❑ We flatten and add a feed-forward neural net for further accuracy.



# Final CNN Product

- ❑ We combine multiple convolutional layers and pooling layers
- ❑ We just set the size of the kernels
- ❑ We let the network optimize values of the filters
- ❑ We flatten and add a feed-forward neural net for further accuracy.

